

ITSC 486 MULTIMEDIA DATA TECHNOLOGIES

Chutisant Kerdvibulvech, Ph.D.

Assistant Professor

ผศ.ดร.ชุตินันต์ เกดวิบูลย์เวช

INSTRUCTOR

Asst. Prof. Dr. Chutisant Kerdvibulvech

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Graduate School of Communication Arts and Management Innovation
National Institute of Development Administration (NIDA)

- Email: chutisant.ker@nida.ac.th
- Website: www.gscm.nida.ac.th/~chutisant/

Education



www.chula.ac.th/

- B.E. (Hons) in Computer Engineering
Chulalongkorn University



www.keio.ac.jp

- MSc. / Ph.D. in Engineering, Department of
Information and Computer Science

Keio University, the oldest university in Japan

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3

Academic Work Experience

- Teaching and being as a research committee in following universities, such as
 - Rangsit University, Sripatum University, PIM
 - Chulalongkorn University, Mahidol University, Kasetsart University
 - NIDA
- Technical Committees for many academic international conferences and journals (Scopus and Scimago)

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4

Media Work Experience (1/3)

1. Newspapers

- IT Columnist for Dailynews newspaper, every Wednesday (page 23)



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5

Media Work Experience (2/3)

2. Magazines

- Columnist for Manager

360° Magazine, iTech 360°

- Columnist for e-commerce Magazine, Augmented Reality



นิตยสารผู้จัดการ 360 องศา พฤษภาคม 2555

จาก Facebook มาสู่ GREE สร้างอาณาจักรเศรษฐกิจที่อายุน้อยที่สุดในโลก
โดย ดร.ชุตินันต์ เกตุวิบูลย์

นิตยสารผู้จัดการ 360 องศา เมษายน 2555

2G จากอดีต มาสู่ 3G ปัจจุบัน ก้าวเข้าสู่ 4G แห่งอนาคต
โดย ดร.ชุตินันต์ เกตุวิบูลย์



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6

Media Work Experience (3/3)

3. Televisions

- IT News Analyst for Nation Channel TV Program



- IT News Analyst for RSU TV Channel



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7

Evaluation

(Technique-based, Creativity-based, and Research-based)

- Project I / Assignments, Participation and Attendance 25 %
- Project II / Assignments, Participation and Attendance 25 %
- Final Project / Assignments, Participation and Attendance 30%

Class Attendance: 10%

Rinkou: 10%

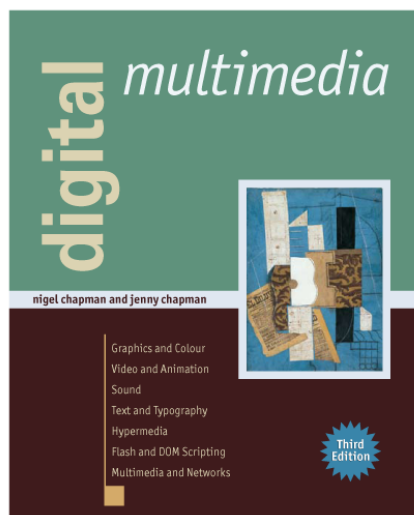
8

Course Syllabus and Lectures

All course syllabus and lectures can be downloaded online via the following links.

- Course Syllabus
 - <http://gscm.nida.ac.th/~chutisant/Multimedia/CS.pdf>
- Lectures
 - <http://gscm.nida.ac.th/~chutisant/Multimedia/L1.pdf>
 - <http://gscm.nida.ac.th/~chutisant/Multimedia/L2.pdf>
 - <http://gscm.nida.ac.th/~chutisant/Multimedia/L3.pdf>
 - etc

9



1

Introduction

**Based on material from
Digital Multimedia, 3rd edition
 published by John Wiley & Sons, 2009
 © 2009 Nigel Chapman and Jenny Chapman**

**These lecture slides © 2009
 Nigel Chapman and Jenny Chapman**

All figures © MacAvon Media Productions

Additional Textbooks

- **Augmented Reality: A Modern Vision-Based Approach, Author: Chutisant Kerdvibulvech, 2nd Edition (2016)**
- **สื่อดิจิทัลใหม่...สื่อแห่งอนาคต, ชุติสันต์ เกิดวิบูลย์เวช คณะนิเทศศาสตร์ และนวัตกรรมการจัดการ สถาบันบัณฑิตพัฒนบริหารศาสตร์ (นิด้า) 2559**

Related-Academic Papers

- C. Kerdvibulvech, “Vision and Virtual-based Human Computer Interaction Applications for a New Digital Media Visualization”, 23rd International Conference on Computer Graphics, Visualization and Computer Vision 2015 (WSCG), Eurographics Association & ACM SIGGRAPH (Scopus, SCImago and Thomson Reuters ISI), Plzen, Czech Republic, June 8 - 12, 2015.
- C. Kerdvibulvech, “An Innovative Use of Multidisciplinary Applications Between Information Technology and Socially Digital Media for Connecting People”, The Multidisciplinary International Social Networks Conference 2015 (Springer), in Communication of Computer and Information Science (CCIS), (Scopus and SCImago Quartile 4th), Ehime, Japan, September 1 – 3, 2015.

Related-Academic Papers

- C. Kerdvibulvech, “A Methodology for Hand and Fingers Motion Analysis Using Adaptive Probabilistic Models,” EURASIP Journal on Embedded Systems (JES), 9 pages, No. 18, Springer Publisher (Elsevier's Scopus, EiCompendex, INSPEC, SciFinder, SCImago, ACM, DBLP, and DOAJ), 2014.
- C. Kerdvibulvech and K. Yamauchi, “Structural Human Shape Analysis for Modeling and Recognition,” Joint IAPR International Workshops on Structural and Syntactic Pattern Recognition (S+SSPR'14), Lecture Notes in Computer Science (LNCS), vol. 8621, Joensuu, Finland, Springer-Verlag Berlin Heidelberg, pp. 282–290, August 20-22, 2014.

Websites and Links

- <http://ieeexplore.ieee.org/>
- <http://dl.acm.org/>
- <http://link.springer.com/bookseries/558>

- <http://www.dailynews.co.th/>
- <http://www.ecommerce-magazine.com/>
- <http://www.manager.co.th>

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15

1

INTRODUCTION

3

Information can be conveyed in the form of text, still images, Web pages, slideshow presentations, video, sound or interactive tooltips.

1

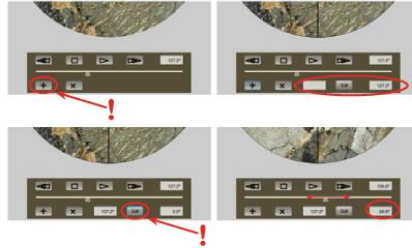
INTRODUCTION

4

Measuring Angles

You can measure the angle between a reference point and some other position using the cross-hairs and the angle read-outs.

1. Click on the button with an upright cross (+) on it to show the cross-hairs. This will cause a new button, labelled Set, to appear, together with two text fields: the one to the left of the Set button (the base angle read-out) will be blank. The other (the angular difference readout) will show a copy of the current angle of rotation.
2. Use the slider or stepping arrows to rotate the slide to the position you want to use as the reference for your measurement.
3. Click the Set button. The current angle will be copied to the base angle readout and the angular difference readout will be set to zero.
4. Use the slider or stepping arrows to rotate the slide to the position where you want to measure the angle.
5. Read the angle in the angular difference readout.



Text

Images

1

INTRODUCTION

5

GeoPlayer Manual

GeoPlayer Home
Contents
Useful Links
Community

Measuring Angles

You can measure the angle between a reference point and some other position using the cross-hairs and the angle readouts.

Click on the button with an upright cross (+) on it to show the cross-hairs. This will cause a new button, labelled Set, to appear, together with two text fields: the one to the left of the Set button (the base angle read-out) will be blank. The other (the angular difference readout) will show a copy of the current angle of rotation.

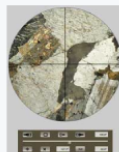
Use the [slider](#) or [stepping arrows](#) to rotate the slide to the position you want to use as the reference for your measurement.

Click the Set button. The current angle will be copied to the base angle readout and the angular difference readout will be set to zero.

Use the [slider](#) or [stepping arrows](#) to rotate the slide to the position where you want to measure the angle.

Read the angle in the angular difference readout.

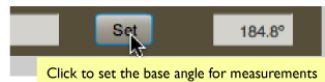
[< previous](#) [next >](#)



[+ enlarge](#)



Video



A Web page

A tool tip

1

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6

There is a fundamental distinction between time-based and static media: time-based media exhibit change over time; static media do not.

1

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7

Video, animation and sound are time-based media.

Still images and text are usually considered to be static media.

1

INTRODUCTION

8

**Each medium has its own characteristics,
leading to distinctive strengths and
weaknesses.**

**Always choose the most appropriate medium
for your purpose.**

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INTRODUCTION

9

Media may be combined into multimedia.

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INTRODUCTION

10

Digital media can be manipulated as data by programs.

1

INTRODUCTION

11

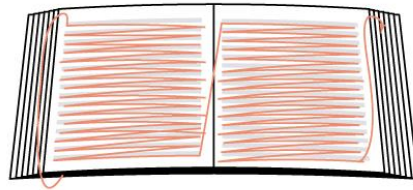
Users can interact with digital multimedia in novel ways, leading to non-linear structures.

1

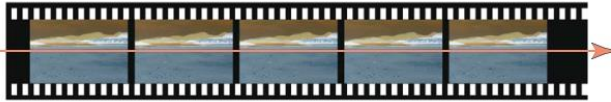
INTRODUCTION

12

Book: physical arrangement of text and pages implies a linear reading order.



Film: fixed order of frames defines a single playback sequence.

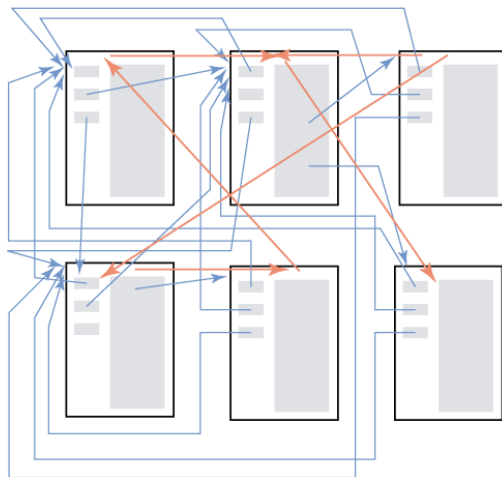


Linear structures in conventional media

1

INTRODUCTION

13



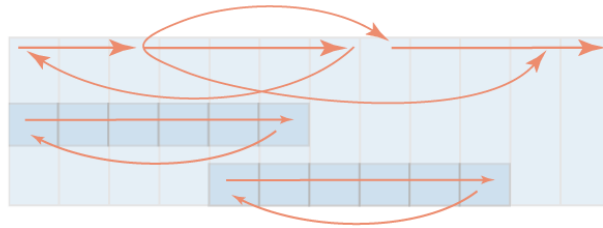
Hypermedia: links between pages permit multiple arbitrary reading orders.

Non-linear structures (1)

1

INTRODUCTION

14



Flash: jumps between frames controlled by interactivity; independent movie clips play in parallel.

Non-linear structures (2)

1

INTRODUCTION

15

Digital multimedia can interact with other sorts of data and computation, serving as a user interface to databases and applications.

1

INTRODUCTION

16

Multimedia is a relatively immature technology, although its adoption is accelerating with the increasing power of computer systems.

1

INTRODUCTION

17

The history of the development of film demonstrates that it takes much more time than multimedia has existed for new media forms to develop fully.

1

INTRODUCTION

18

Most multimedia adapts the forms of older media, but unique new forms are beginning to emerge.

Virtual Reality



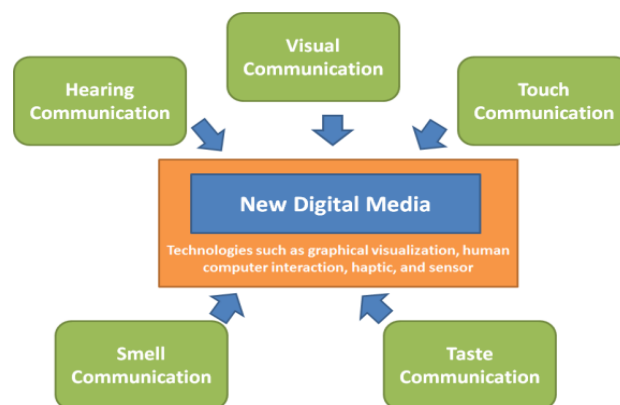
"The Oculus community is leading us into the future, and we can't wait for everyone to experience the magic of VR and the next platform."...Mark Zuckerberg, CEO of Facebook

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24 September 2015

New Digital Media

New Digital Media



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Visual Communication

Augmented Reality



Augmented Reality Magic
(card trick), Newmagic
Communications Inc.

TED
TALKS

Sixth Sense (MIT)

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Mixed Reality for Military Application



Dr. Adrian Cheok,
National University of Singapore (NUS)

Broadcast in CNN



4



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GSCM Digital Campaign (AR)



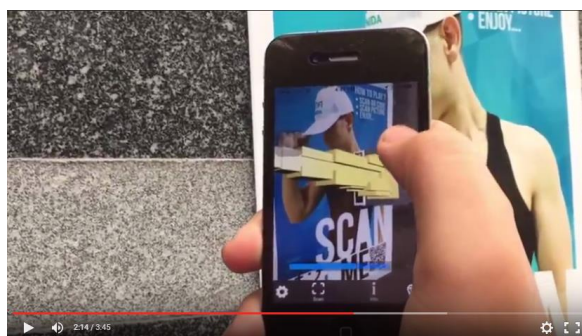
Pixlive Maker

Pixlive Player

**PIXLIVE
MAKER**
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<http://www.vidinoti.com/>



Next Poster



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NIDA VirtualBox



Touch Communication

Huggable Robot (MIT Media Lab)



<http://robotic.media.mit.edu/portfolio/huggable/>

A blue fancy robotic bear was designed to relieve pain for kids in the Wired Magazine

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Touch Communication

Huggy Pajama (Keio)



<http://www.sciencechannel.com/tv-shows/pop-scis-future-of/videos/popscis-future-of-huggy-pajamas/>
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Smell Communication

A smell-based system called Dollars & Scents inside the entrance to their laboratory at MIT

- The system releases scents and perfumes into the air differently depending on stock market changes.
- Emit a lemon-like fragrance if NASDAQ index is going down, while they emit a rose-like fragrance if the stock market is going up.
- Similarly, they also built an olfaction system, called inStink, for conveying ambient presence and activity awareness for food.

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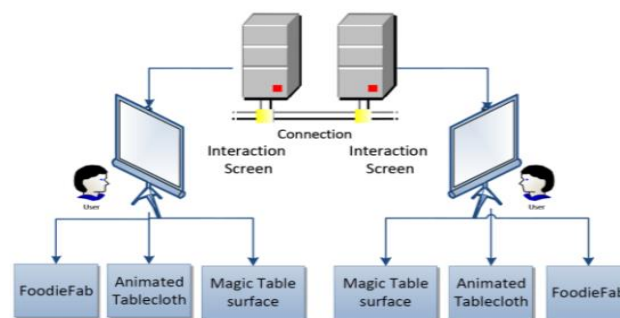
Kissenger

- Kissenger (National Taipei University)
- Kiss Messenger
- Kiss Transmission Robot



Dr. Hooman Samani, College of Electrical Engineering and Computer Science, NTU

Taste Communication



The Food Media's system configuration is set up by [Wei13a] from National University of Singapore.